WORK SHEET - 07 **COMPLEX NUMBER**

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.11 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Q.1 The complex number z which satisfies the equations $|z| = 1$ and $\left|\dfrac{z-\sqrt{2}(1+i)}{z}\right| = 1$ is

(A) 1 (B) $1+i$ (C) $\dfrac{1+i}{\sqrt{2}}$ (D) $\dfrac{-1-i}{\sqrt{2}}$

Q.2 If $z \in C$ then area enclosed by the curve $2\left|z^2+|z|^2\right| + 3\left|z^2-|z|^2\right| = 6|z|,\ z \neq 0$ is

(A) 3 sq. units (B) 6 sq. units (C) 3π sq. units (D) 6π sq. unit

Q.3 The complex numbers z_1, z_2 satisfies $\dfrac{z_2}{z_1} = \dfrac{1+i\sqrt{3}}{2}$, then origin, z_1 and z_2 are the vertices of a triangle which is

(A) of area zero (B) right-angled isosceles
(C) equilateral (D) obtuse-angled isosceles

Q.4 Let $z = \cos\theta + i\sin\theta$, then the value of $\sum\limits_{n=0}^{9} \text{Re}\left(\dfrac{4i}{z^{2n+1}}\right)$ at $\theta = 3°$ is equal to

[**Note :** Re (z) denotes real part of z and $i = \sqrt{-1}$.]

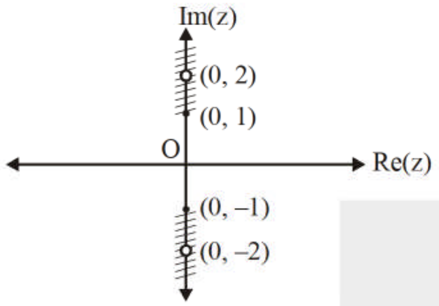
(A) $\dfrac{1}{2}$ cosec 3° (B) $\dfrac{1}{4}$ cosec 3° (C) cosec 3° (D) $\dfrac{1}{10}$ cosec 3°

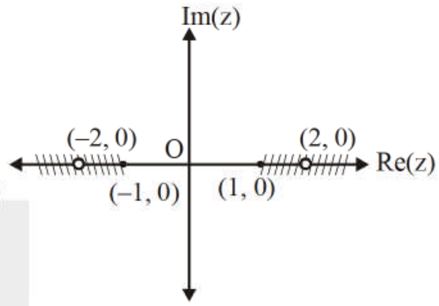
Q.5 Let $z = x + iy$ be a complex number where $x, y \in R$ and $i = \sqrt{-1}$. The area bounded by locus of $P(z)$ satisfying $|z-1| = 2\,I_m(z)$ and coordinate axes on the complex plane, is equal to

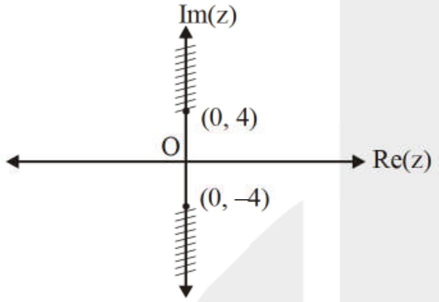
(A) $\sqrt{3}$ (B) $2\sqrt{3}$ (C) $\dfrac{1}{\sqrt{3}}$ (D) $\dfrac{1}{2\sqrt{3}}$

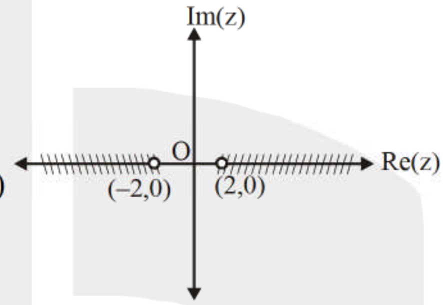
[**Note :** $I_m(z)$ denotes imaginary part of z.]

Q.6 The set $\left\{\left(\dfrac{-4iz}{4-z^2}\right) : \text{z is a complex number}, |z| = 2, z \neq \pm 2\right\}$ is best represented by

(A)


(B)


(C)


(D)


Q.7 The area enclosed by the curves $C_1 : \arg(z-2) = \dfrac{\pi}{4}$, $C_2 : \arg(z-2) = \dfrac{3\pi}{4}$ and $C_3 : (\text{Re}(z-2))^2 = \left(\dfrac{z-\bar{z}}{2i}\right)$ on the complex plane is

(A) $\dfrac{2}{3}$ (B) $\dfrac{1}{3}$ (C) $\dfrac{5}{6}$ (D) $\dfrac{4}{3}$

Q.8 The number of complex numbers z satisfying simultaneously $|z+1-i| = 2$ and $\text{Re}(z) \geq 1$ equals
(A) 0 (B) 1
(C) 2 (D) infinite

Q.9 The complex number z satisfies the condition $\left|z - \dfrac{25}{z}\right| = 24$. The maximum distance from the origin of co-ordinates to the point z is
(A) 25 (B) 30 (C) 32 (D) 16

Q.10 If $z \in C$ and satisfies the equation $(z-\bar{z})^2 = 4|z|^2 - 12$ then the maximum value of $|z|$ is

(A) $\sqrt{6}$ (B) $2\sqrt{3}$ (C) $\sqrt{3}$ (D) $\dfrac{\sqrt{3}}{2}$

Q.11 Let z_1 and z_2 be nth roots of unity which subtend a right angle at the origin. Then n must be of the form
(A) 4k + 1 (B) 4k + 2 (C) 4k + 3 (D) 4k

[PARAGRAPH TYPE]

Q.12 to Q.14 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 12 to 14

Let two curves $C_1 : x^2 + y^2 = 2$ and C_2 : locus of z which satisfies $\left||z+3\sqrt{2}|-|z-3\sqrt{2}|\right| = 2\sqrt{2}$.

Q.12 If z_0 lies on C_1 then maximum value of $|z_0 + 4 + 4i|$ is

(A) $5\sqrt{2}$ (B) $6\sqrt{2}$ (C) $8\sqrt{2}$ (D) $16\sqrt{2}$

Q.13 If $z_1\left(\text{Re}(z_1) > 0\right)$ lies on C_1 and C_2 both and z_2 is obtained by rotating z_1 about origin in anticlockwise direction through $135°$ and $z_3 = \bar{z}_2$ then real part of $(z_1 + z_2 + z_3)$ equals

(A) $-2-\sqrt{2}$ (B) $-\sqrt{2}+1$ (C) $-2+\sqrt{2}$ (D) $-\sqrt{2}-1$

Q.14 If locus of z satisfying $|\arg(z-1)| = \tan^{-1}(4)$ meets the curve C_2 at A and B then area of the triangle formed by A, B and C where complex number corresponding to C is $e^{i\,2\pi}$, is

(A) 4 sq. units (B) $8\sqrt{2}$ sq. units (C) 16 sq. units (D) $16\sqrt{2}$ sq. units

[MULTIPLE CORRECT CHOICE TYPE]

Q.15 to Q.16 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

Q.15 Let z be a variable complex number satisfying $\dfrac{2z-\bar{z}}{3} = a - \dfrac{1}{3} + ib$ where $a, b \in R$ and $a^2 - b^2 = 1$.

If locus of z is the curve 'C' then

(A) director circle of the curve 'C' is $(x+1)^2 + y^2 = 10$.

(B) area of the triangle formed by the straight line $x - 3y + 1 = 0$ and tangent at one of the vertices and the x-axis is $\dfrac{3}{2}$ sq. units.

(C) distance between directrices of the curve 'C' is $\dfrac{18}{\sqrt{10}}$.

(D) equation of the circle passing through the points (2, 1) and both foci of the curve 'C' is $x^2 + y^2 + 2x - 9 = 0$.

Q.16 Let P(z) satisfies $z\bar{z} + (4-5i)\bar{z} + (4+5i)z = 40$. If $a = \max.|z+2-3i|$ and $b = \min.|z+2-3i|$, then

(A) $a + b = 18$ (B) $a + b = 9$ (C) $a - b = 4\sqrt{2}$ (D) $ab = 73$

[INTEGER TYPE]

Q.17 to Q.20 are "Integer Type" questions. (The answer to each of the questions **are upto <u>4 digits</u>**)

Q.17 Consider the circle $|z - 5i| = 3$ and two points z_1 and z_2 on it are such that $|z_1| < |z_2|$ and $\arg. z_1 = \arg. z_2 = \frac{\pi}{3}$. A tangent is drawn at z_2 to the circle, which cuts the real axis at z_3, then find $|z_3|^2$.

Q.18 Consider $\alpha_k = \cos\frac{2k\pi}{13} + i\sin\frac{2k\pi}{13}$, $k = 0, 1, 2, \ldots\ldots, 12$, if $S = \frac{\sum_{k=1}^{12}\left|\overline{\alpha}_k + \alpha_{12-k}\right|}{\sum_{k=1}^{6}\left|\alpha_{2k-1} + \alpha_{2k}\right|}$, then find the value of S.

Q.19 Let $A(z_1)$ be the point of intersection of curves $\arg(z - 2 + i) = \frac{3\pi}{4}$ and $\arg(z + \sqrt{3}\,i) = \frac{\pi}{3}$, $B(z_2)$ be the point on $\arg(z + \sqrt{3}\,i) = \frac{\pi}{3}$ such that $|z_2 - 5|$ is minimum and $C(z_3)$ be the centre of circle $|z - 5| = 3$. If Δ denotes area of triangle ABC, then find the value of Δ^2.

Q.20 Let $P(z) = z^3 + az^2 + bz + c$ where a, b and c are real. There exists a complex number ω such that the three roots of $P(z)$ are $\omega + 3i$, $\omega + 9i$ and $2\omega - 4$ where $i^2 = -1$. Find the value of $|a + b + c|$.

ANSWER KEY

Q.1	C	Q.2	A	Q.3	C	Q.4	C	Q.5	D
Q.6	B	Q.7	B	Q.8	B	Q.9	A	Q.10	C
Q.11	D	Q.12	A	Q.13	C	Q.14	A	Q.15	BCD
Q.16	ACD	Q.17	[11]	Q.18	[2]	Q.19	[12]	Q.20	[136]